Explore or Exploit: The Mathematics Of Learning From Feedback

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Two Conflicting Objectives

Every video you watch has two jobs:

- Help you learn the subject
- Help you learn which teacher is best

Exploration versus Exploitation

- **Exploration:** Trying new things to learn
- **Exploitation:** Use what currently seems best



Two Ways To Learn From Feedback



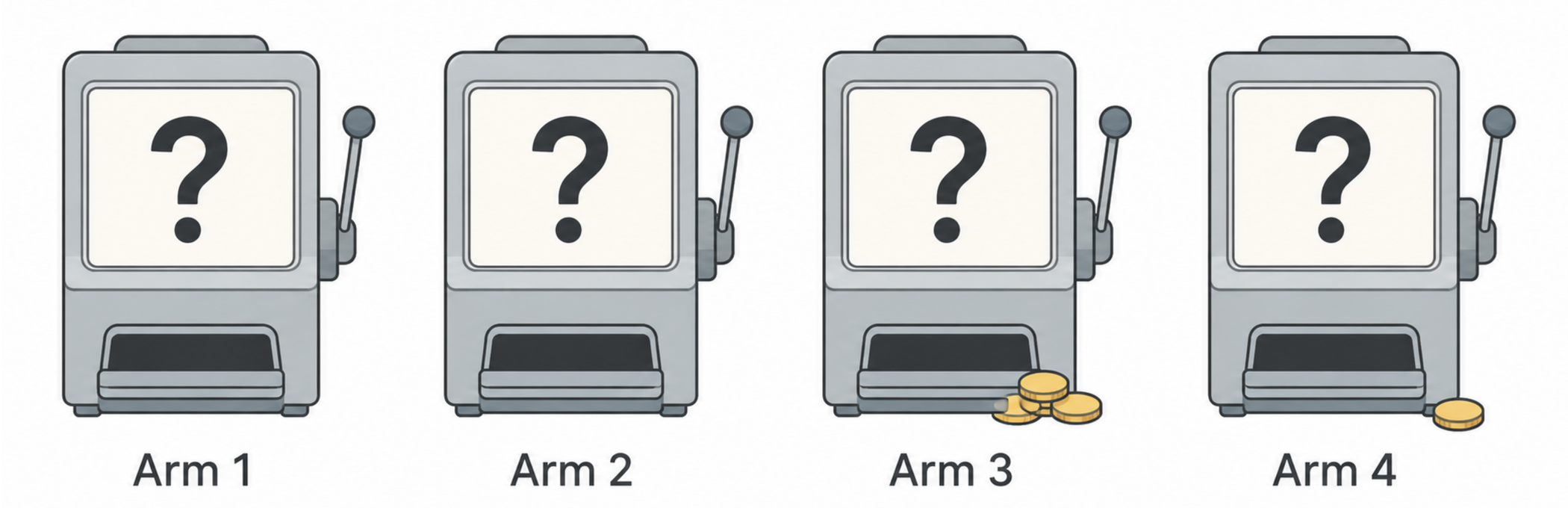
Offline Learning

Online Learning

How do we learn what's best when every choice we make changes what we can learn?

This is called the ***multi-armed bandit*** problem

Multi-Armed Bandit

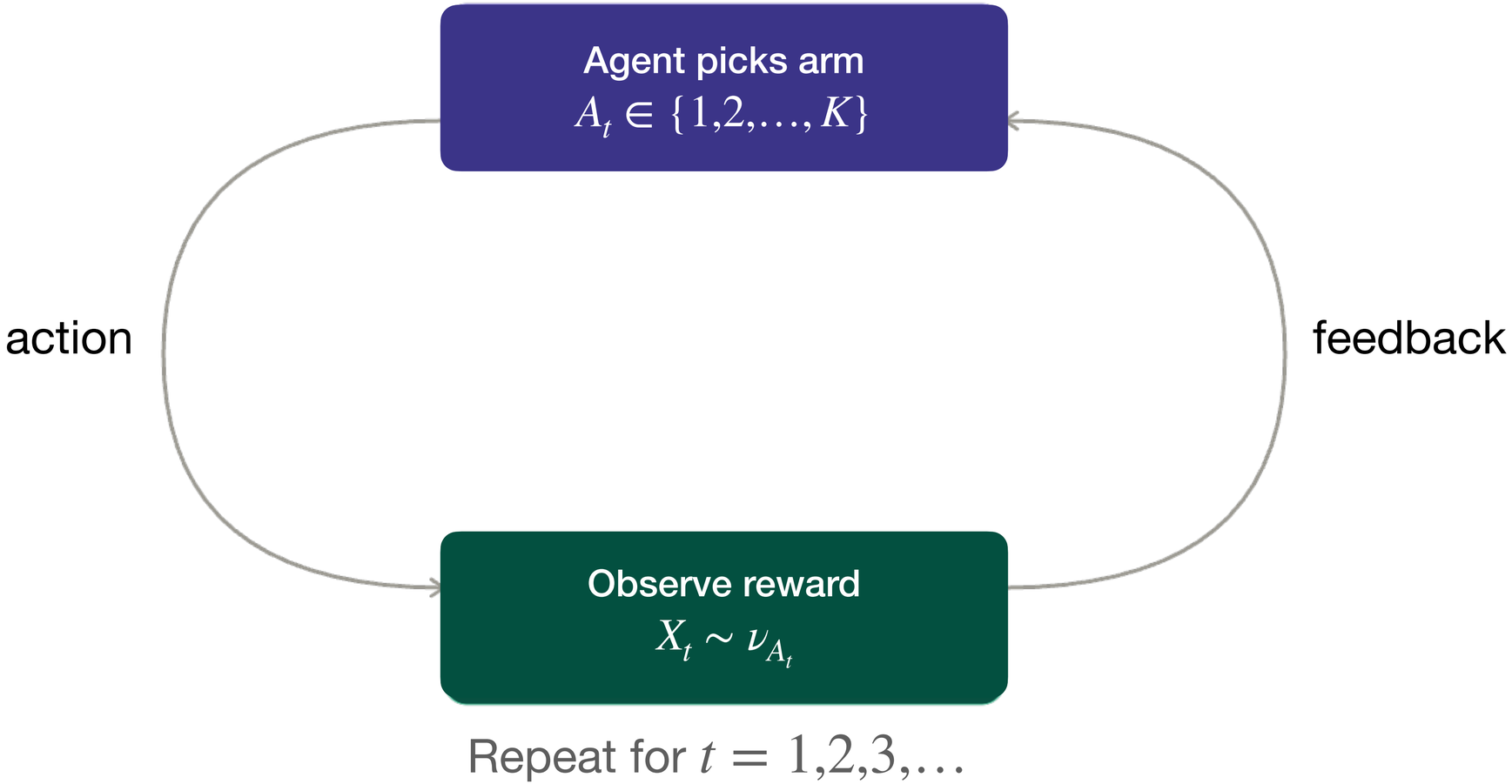


K slot machines (bandits), where pulling an arm yields a random reward each time

Each arm i generates rewards with mean (average) μ_i

Multi-Armed Bandit

The Interaction Protocol



Translation

Video Recommendation	Bandit Formalism
K channels	K arms
Time spent on one video	Random reward X_t
Best channel	i with maximum μ_i
Overall engagement	Cumulative reward $\sum_t X_t$

The Goal

Maximise total reward: $\sum_{t=1}^{T} X_t$

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we would pull i^* every time

The Goal

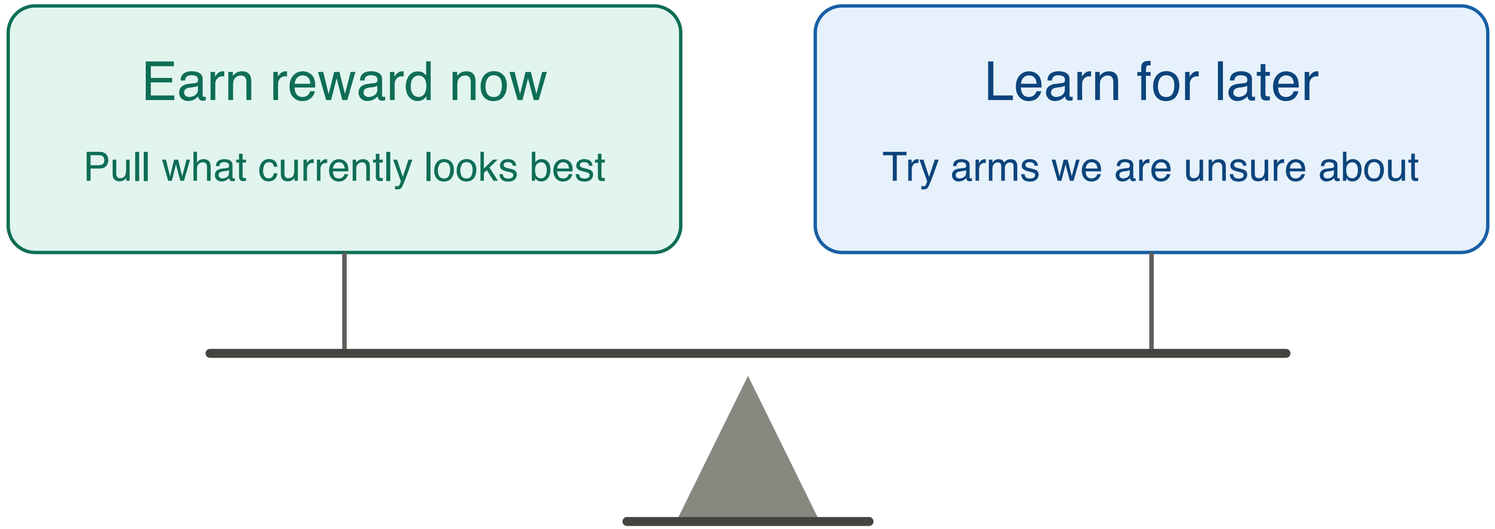
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But we do not know i^*

The Fundamental Difficulty



A good policy must do both

Every recommendation is also an experiment

Which Channel Would You Watch?

Channel	Average Rating	Times Watched
1	9/10	5
2	8/10	3
3	7/10	20
4	4/10	1

Estimating The Unknown

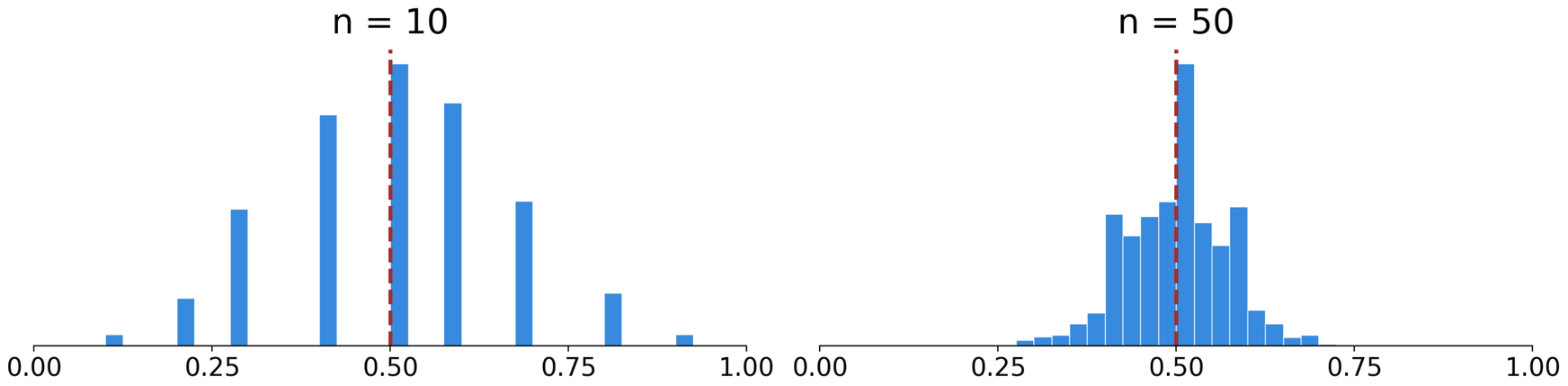
If an arm is pulled n times, we estimate its mean by $\hat{\mu} = \dfrac{X_1 + \ldots + X_n}{n}$

But an average based on $n = 10$ is not as reliable as that based on $n = 50$

Estimating The Unknown

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If we repeated the experiment many times, $\hat{\mu}$ would be spread like this

Quantifying Uncertainty

Hoeffding's Inequality: for arms with rewards in $[0,1]$

After pulling arm i exactly T_i times

$$\Pr\left(|\hat{\mu}_i - \mu_i| \geq \epsilon \right) \leq 2\exp(-2T_i\epsilon^2)$$

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Averages become reliable as the number of samples grows

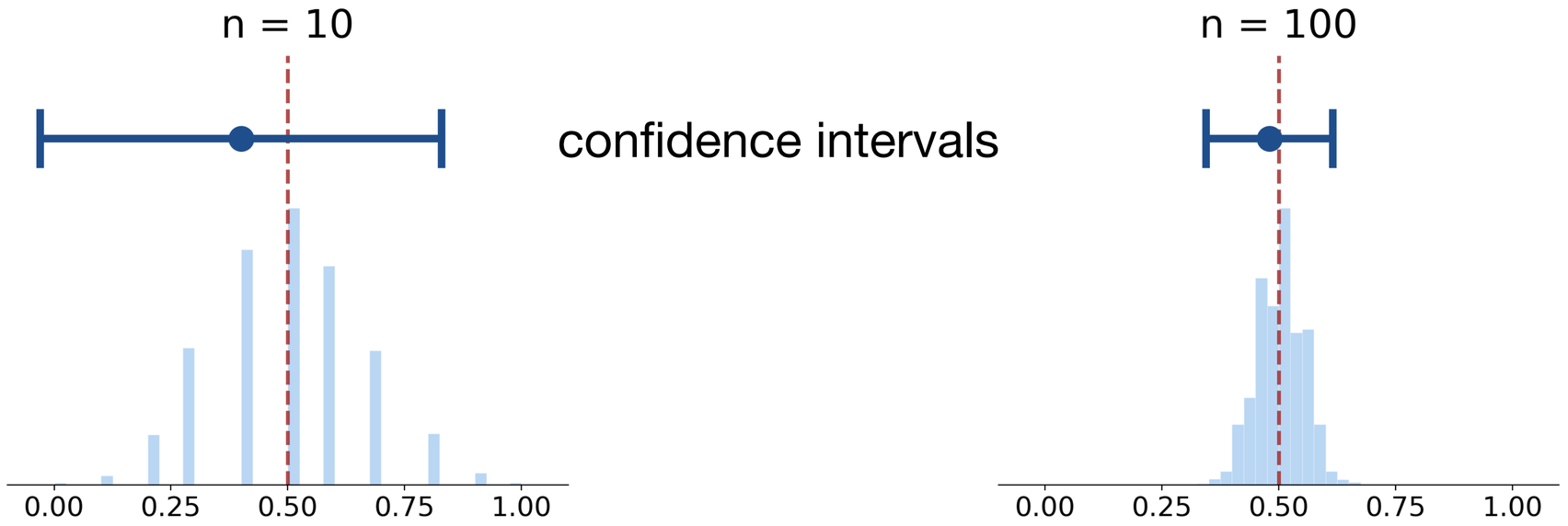
$$T_i \uparrow \Rightarrow \text{typical error } |\hat{\mu} - \mu| \downarrow$$

From Hoeffding to Confidence Intervals

With high probability, $\mu_i \in \left[\hat{\mu}_i(t) - \sqrt{\frac{c}{T_i(t)}},\ \hat{\mu}_i(t) + \sqrt{\frac{c}{T_i(t)}}\right]$

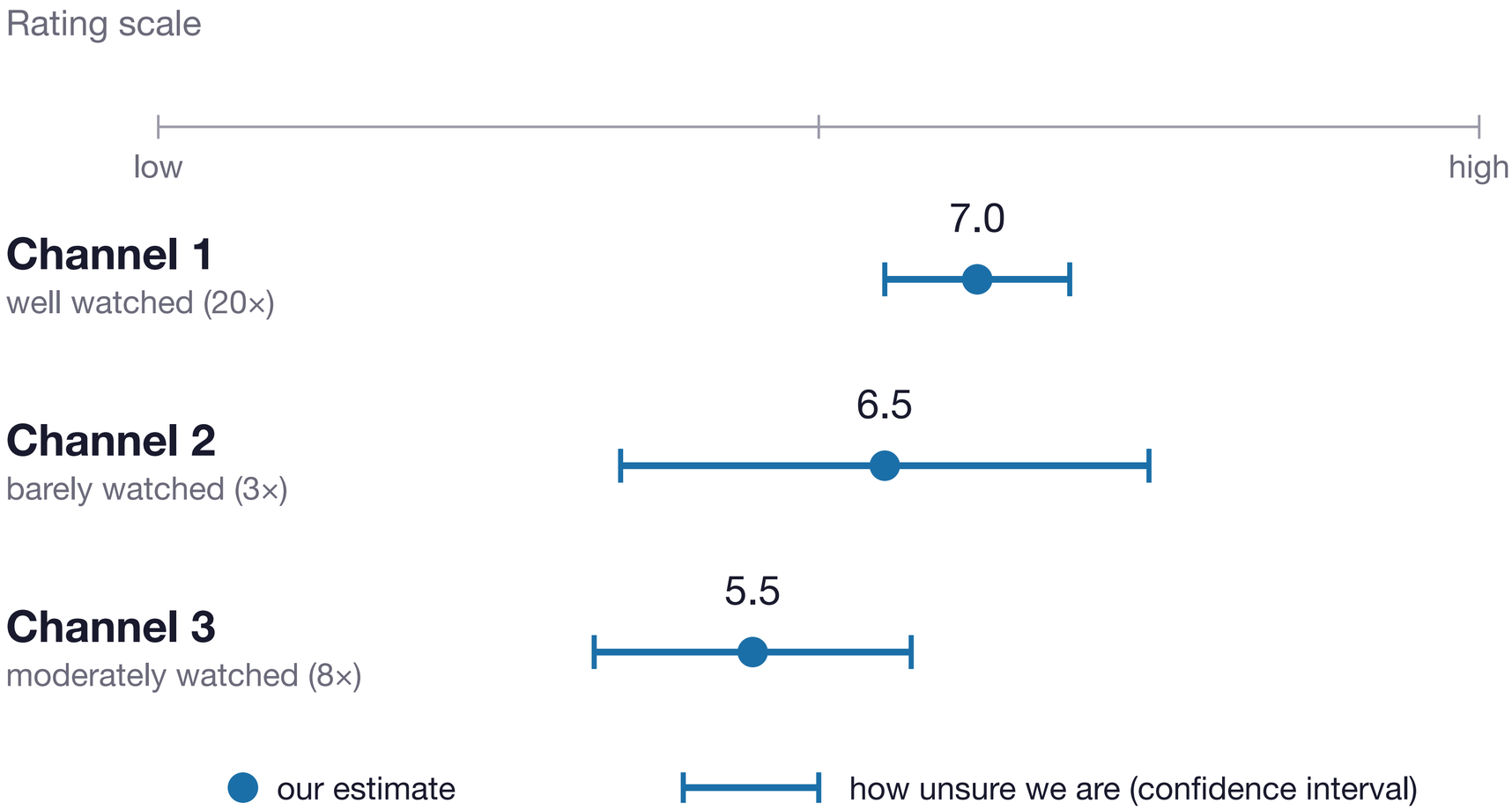
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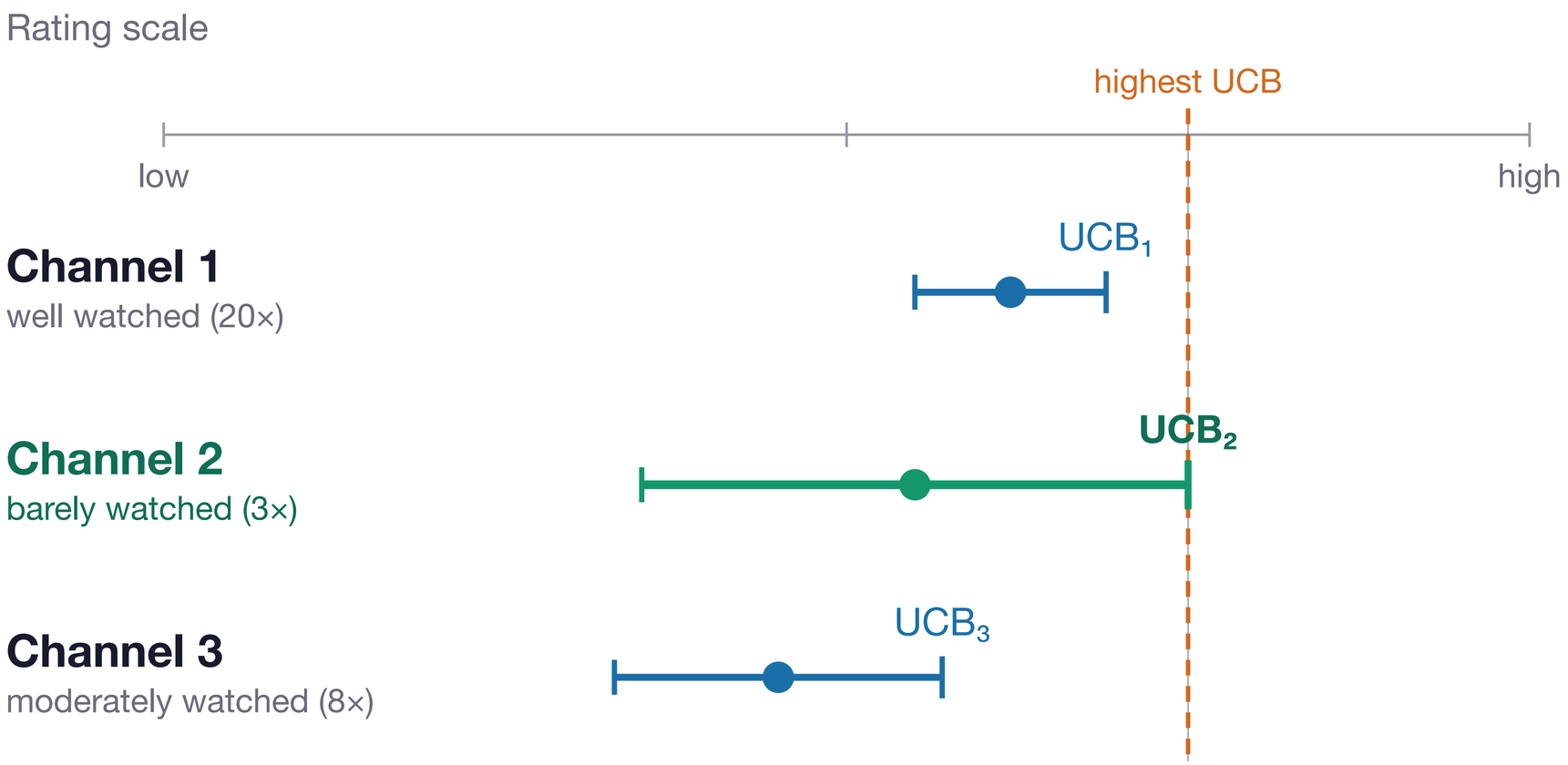


More data ⇒ Narrower intervals

The State of the System



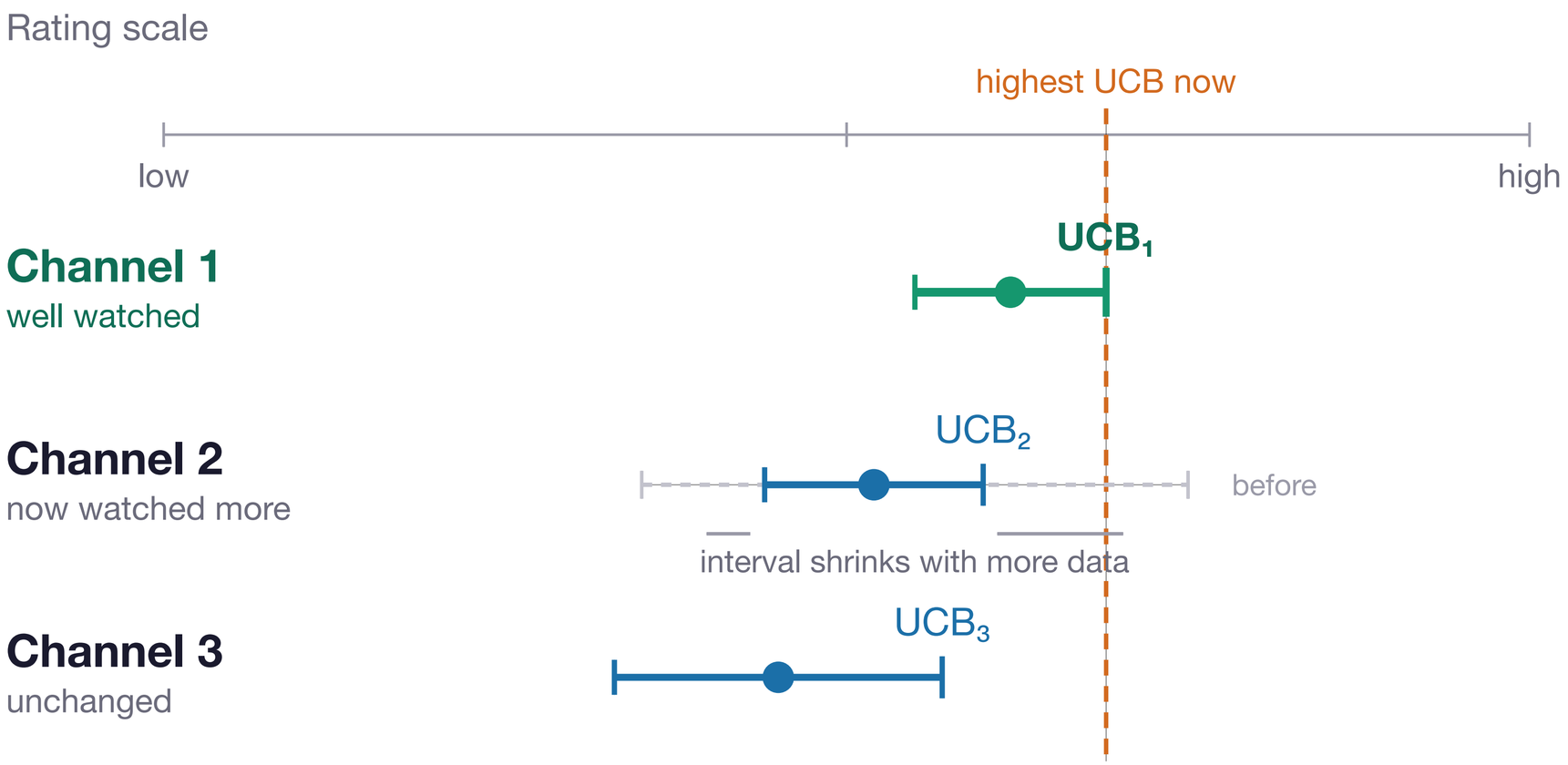
Optimism in the Face of Uncertainty



Channel 2 has the highest UCB — so we watch it next

Always pick the highest UCB — this is the UCB algorithm

UCB’s Self-Correcting Mechanism



Channel 2's UCB dropped — now Channel 1 wins, and we settle there

A bad channel can only fool us for so long

Back to Your Feed

- Every recommendation is also an experiment
- Clicks, skips, watch-time → feedback
- More feedback → less uncertainty about you
- Real systems are much bigger, but the same tension remains:

explore vs exploit



My Research

Recommender Systems That Learn From Comparisons

Absolute rating



Orbit Academy



Hard to score exactly

Pairwise comparison

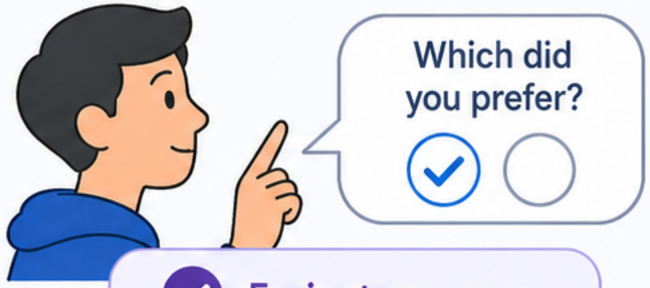


Orbit Academy

VS

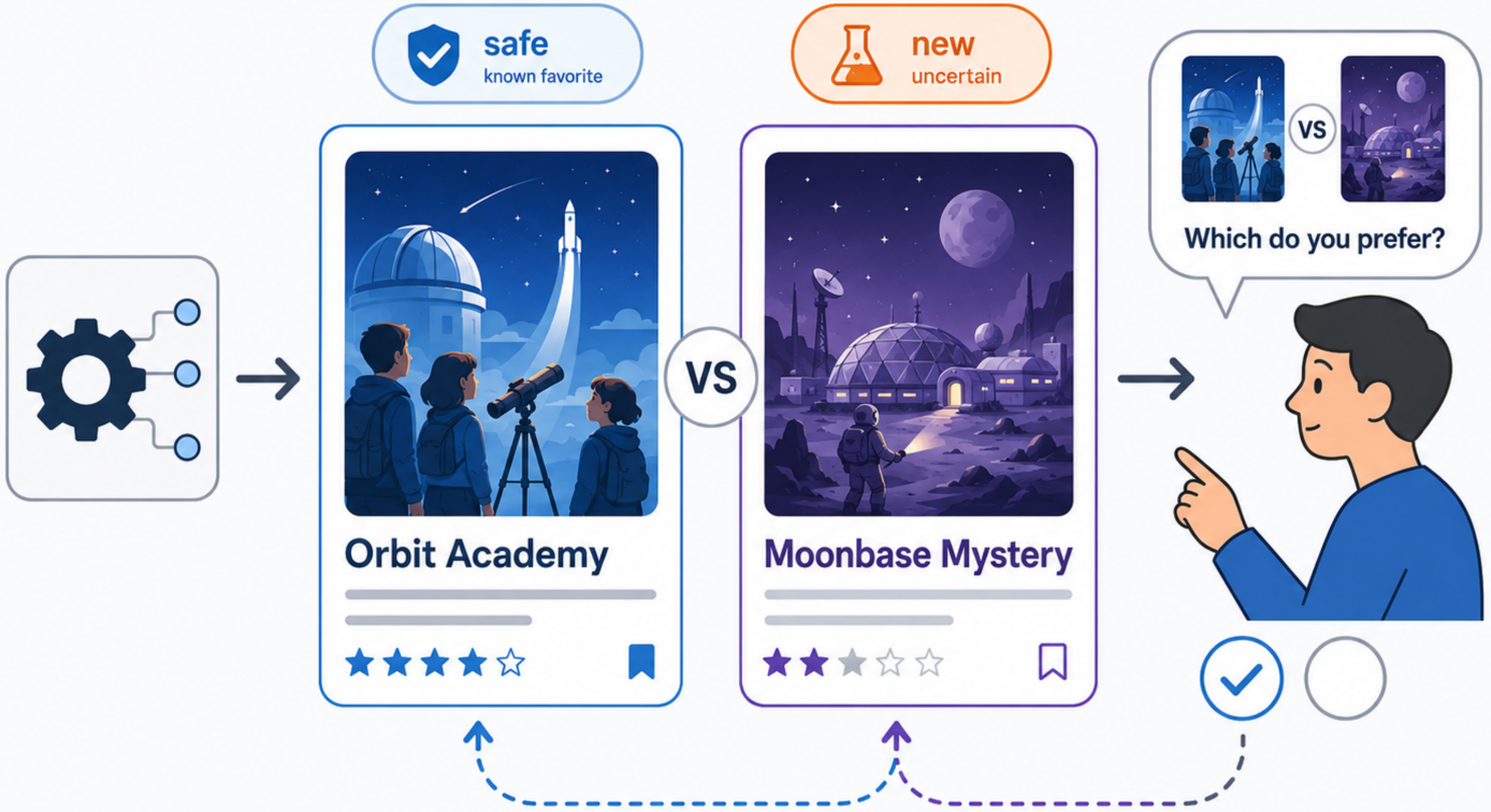


Moonbase Mystery



Easier to compare

Comparisons reveal preferences more directly.



Comparisons can let us exploit with one item and explore with another.

“To teach how to live without certainty, and yet without being paralyzed by hesitation, is perhaps the chief thing that philosophy, in our age, can still do for those who study it.”

–Bertrand Russell

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“To teach how to choose without certainty, and yet without spending your whole life exploring, is perhaps the chief thing that bandit algorithms can do for recommender systems.”

–ChatGPT